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Paleomagnetic investigation of the Early Permian Panjal Traps of NW India; regional tectonic implications

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ABSTRACT

The ~289 Ma Panjal Traps of NW India (Kashmir) are part of a series of rift-related mafic suites (Abor, Sikkim etc.) that were erupted onto northern India (present-day coordinates) around the same time as separation of the Cimmerian blocks of Qiangtang and Sibumasu. We report new data from only the second paleomagnetic investigation of this unit. Standard alternating field and thermal demagnetization methods were used to isolate characteristic magnetizations from seven outcrops at three locations within the Kashmir Valley, NW India. Analysis of four sections (14 individual cooling units) from close to Srinagar, that together form a tectonically coherent sequence spanning ~3 km of stratigraphy, yield a single-component, primary magnetization with a mean direction of Dec: 134.8°, Inc: 55.3° (α95 = 8.9°, k = 21.0). An inclination-only mean of 52.5° (α95 = 8.9°, k = 47.2) gives a paleolatitude of ~33°S (±5°). A paleopole of 110.5°E 8.4°S (A95 = 10.7) is also calculated. Assuming the magnetization records a portion of the reverse polarity Kiaman Superchron, the new result indicates extrusion of the Panjal Traps basalts at mid-latitudes in the southern hemisphere. By inference this constrains the location of central Gondwana, and informs debates related to Cimmeria's detachment from Gondwana.

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1. Introduction

The mid-Early Permian represents an important phase in Pangaea's development marking the time when the >13,000 km-long string of terranes (northern Africa–NE Arabia–northern India–NW and northern Australia) collectively known as 'Cimmeria' separated from Gondwana's Tethyan margin (Sengör, 1979, 1987; Şengör et al., 1988). The use of paleomagnetic data from rift-related flood basalts that formed at this time provides a means to independently constrain the location of India's northern margin. As such, the data inform understanding of the configuration of the eastern Cimmerian terrane along the northern Gondwana margin.

Today, the most spectacular example of continental collision and orogenic processes is the Himalayan mountain range. It is a consequence of continuing convergence between the Indian and Eurasian plates that are now joined along the Indus–Yarlung–Tsangpo Suture (IYTS). The timing and nature of this collision is a matter of on-going debate with estimates ranging from ~65 Ma to as late as 35 Ma (Searle et al., 1987; Hodges, 2000; Aitchison et al., 2007; Ali and Aitchison, 2008, 2012; Aitchison and Ali, 2012; van Hinsbergen et al., 2012a; Bouilhol et al., 2013). Inextricably linked to this debate, the size of the pre-collisional India or 'Greater India' is the center of a similar scale controversy (Ali and Aitchison, 2005; van Hinsbergen et al., 2012a,b; Gibbons et al., 2012). The extent of Greater India has implications not only for the timing of collision with Eurasia in the Cenozoic but for our understanding of how the sub-continent developed through its deeper history.

2. Panjal Traps and geology of the study area

The northern margin of India includes Late Paleozoic to Mesozoic Tethyan passive margin sequences, which today form part of the Higher Himalaya (Garzanti et al., 1996a,b). During the Late Carboniferous to Early Permian, terrigenous rocks were deposited in response to differential uplift of the Indian margin that was linked to rifting (Vannay and Spring, 1993; Garzanti et al., 1999). This extension was accompanied by eruption of the Panjal Traps, Abor

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volcanics, Nar-Tsum spilites and Bhote Kosi basalts of the eastern and central Himalaya and eventually led to the formation of the Neotethyan Ocean and the ribbon-like continent Cimmeria (Acharyya, 1980; Bhat et al., 1981; Bhat, 1984; Sengör, 1987; Garzanti et al., 1999; Shellnutt et al., 2011; Ali et al., 2012).

The Panjal Traps (Lydekker, 1883) are a suite of predominantly mafic volcanic rocks that were erupted onto the Greater India portion of the Neotethyan margin (Bhat and Zainuddin, 1979; Gaetani and Garzanti, 1991). They occur within the Tethyan Himalaya of NW India, with the largest exposures within the Kashmir Valley and on its periphery (Bhat and Zainuddin, 1979) (Fig. 1). The Traps form part of the Pangaea Megasequence, conformably overlying pyroclastic agglomeratic shales and the freshwater, plant-bearing, siliciclastic Nishatbagh beds (Wopfner and Jin, 2009). They are in turn succeeded by the *Gangamopteris*-bearing Mamal Beds that lie directly below the Zewan Formation (Bhat and Zainuddin, 1978; Kapoor and Tokuoka, 1985). Notably, the Permo–Triassic boundary occurs within the lower part of the succeeding Khunamuh Formation (Kapoor and Tokuoka, 1985). The Triassic sequence comprises shallow marine deposits of the Tethyan passive shelf (Bhat and Zainuddin, 1978). Equivalent sections of the sequences overlying the Panjal Traps occur in the Bingda-Yongde Formation redbeds of the Baoshan Block where they overlie basalts of the Woniusi Formation, which are regarded as Panjal correlatives (Wang et al., 2001; Ueno, 2003).

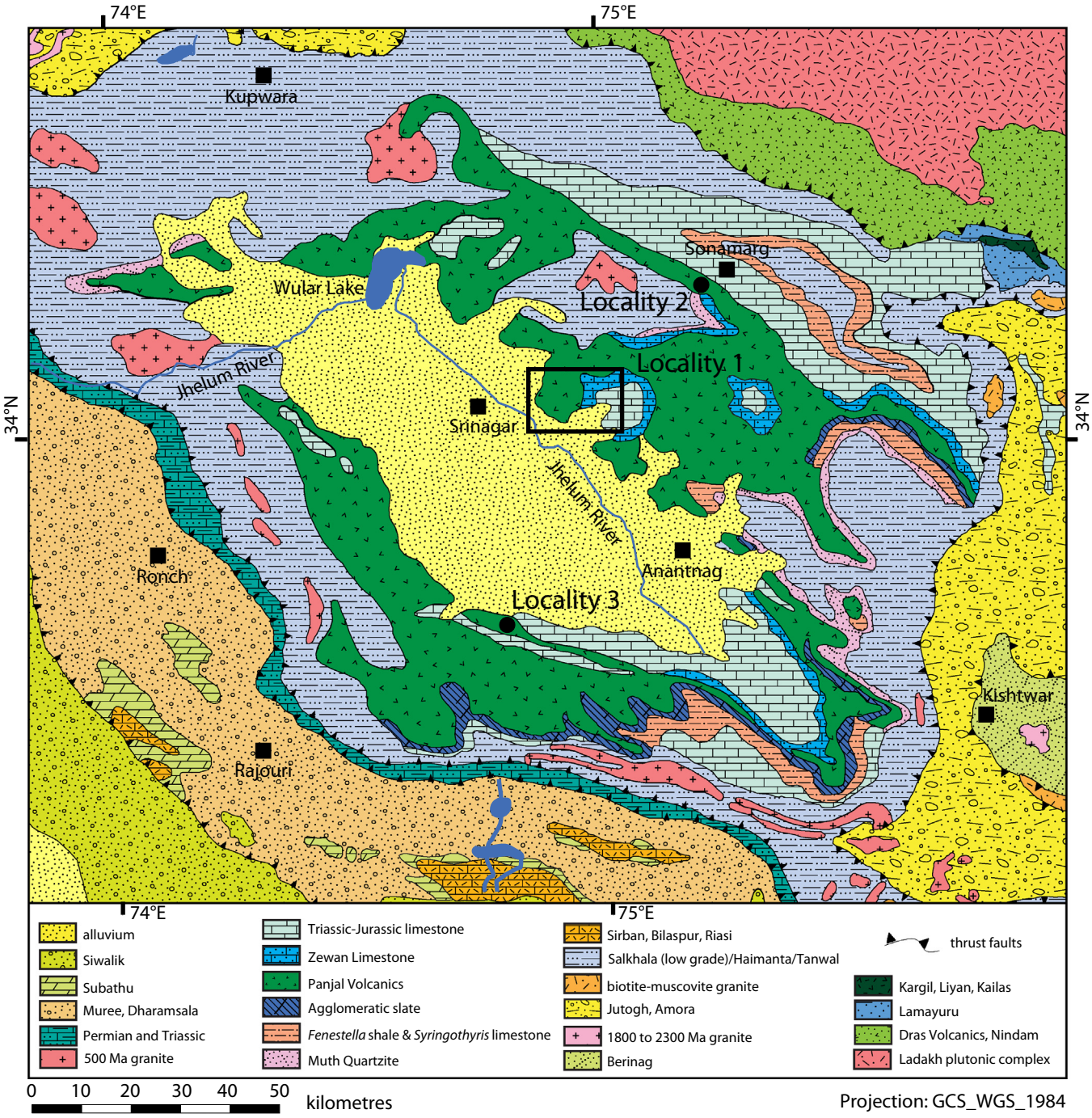


Fig. 1. Location map showing the distribution of geological units in the Kashmir Valley and sample locations (after Dar et al. (2014)).

Remnants of the Panjal Traps crop out from easternmost Zanskar-Spiti (East Kashmir) (Fuchs, 1987; Vannay and Spring, 1993) to areas within the Western Himalayan Syntaxis in NE Pakistan (Papritz and Rey, 1989) and cover a combined area >10^5 km^2. Although their original distribution remains uncertain, it has been suggested that, along with related Permian volcanics in the Himalaya, their extent may have reached 0.2×10^6 km^2 (Ernst et al., 2005). Their scale is thus similar to both the Emeishan Large Igneous Province (ELIP) of SW China and the Columbia River basalts of the NW USA (Ali et al., 2005; Bryan and Ernst, 2008; Shellnutt et al., 2011). The Traps consist mostly of basalts with minor basaltic-andesites, rhyolites and dacites (Ganju, 1944; Nakazawa and Kapoor, 1973; Pareek, 1976; Bhat et al., 1981; Shellnutt et al., 2011, 2012, 2014, 2015). Zircon LA-ICP-MS U–Pb age dating of a rhyolite from the lower-middle volcanic sequence of the Panjal Traps yielded a zircon U/Pb age of 289 ± 3 Ma (Shellnutt et al., 2011). The thickness of the basaltic succession ranges between ~300 km in the Zanskar Range to ~3 km in the Pir Panjal Range, with individual flows up to 30 m thick (Middlemiss, 1910; Wadia, 1934; Fuchs, 1987; Gaetani et al., 1990; Chauvet et al., 2008). Nakazawa and Kapoor (1973) have suggested that the Panjal volcanic center was located in western Kashmir because the total flow thickness is greater than other regions. Most eruptions were subaerial (Wopfner and Jin, 2009) although some exposures around Srinagar (i.e. Kashmir Valley) exhibit pillow structures indicating eruption in a lacustrine setting (Nakazawa et al., 1975; Pareek, 1976; Bhat and Zainuddin, 1979). Silicic volcanic rocks are concentrated within the Kashmir valley beneath the basalt flows and appear to be restricted to the Pir Panjal Range (Ganju, 1944; Shellnutt et al., 2012). The stratigraphic position of the silicic units is somewhat unusual in that they occur as the lower flows whereas in other LIPs they are typically found in the waning sequences (Sheth, 2007).

In terms of the geotectonic setting in which the traps were erupted, Detailed geochemical studies have identified signatures typical of tholeiitic OIB/P-MORB-type basalts consistent with a rifting scenario (Vannay and Spring, 1993; Chauvet et al., 2008).

However, the specific mechanism that initiated the rifting has not been fully resolved. Geological and large-scale paleogeographical relationships suggest a passive rifting scenario resulting from increasing slab-pull forces in the Paleotethys following the subduction of a mid-oceanic ridge beneath the Eurasian margin to the north (Gaetani and Garzanti, 1991; Stampfli et al., 1991; Garzanti et al., 1996a,b; Stampfli and Borel, 2002). An alternative scenario invokes the involvement of a Tethyan mantle plume forming the Arabia–India–Cimmeria triple junction (Lapierre et al., 2004; Chauvet et al., 2009).

The spatio-temporal association between the extrusion of the Panjal Traps and the rifting of the Cimmerian terranes is important for determining the location of the rifting margin. Chauvet et al. (2008) suggested that the Traps erupted during the Middle to Late Permian. More recent U/Pb zircon dating work by Shellnutt et al. (2011) has yielded an Early Permian age of ~289 Ma. Thus, the Traps form the locus of the India–Cimmeria rifting margin and potentially may provide a paleolatitudinal constraint for defining the extent of Greater India.

3. Paleomagnetic investigation

3.1. Previous work

McElhinny et al. (1978) reported paleomagnetic data from thirteen flows collected in the vicinity of Srinagar during a 1967 Geological Survey of India expedition. From twelve of these samples that were subjected to a detailed step-wise demagnetization (alternating field and thermal) they reported a mean magnetization of Dec: 156.5° Inc: 32.5° (α95 = 9.9°, k = 19.8, n = 12), and a resulting paleopole of 32°N 282°E. The samples yielded a positive fold test, suggesting magnetization during Early Tertiary Himalayan uplift McElhinny et al. (1978). They noted similarity between Panjal and the Tertiary Deccan Traps poles (Wensink, 1973) but apparent disparity with other Permian poles from the Salt Range, Pakistan (Wensink, 1975) and the Indian Peninsular shield (Wensink, 1968; Klootwijk, 1975). McElhinny et al. (1978) concluded that their mean magnetization was likely an overprint acquired during the formation of the Himalayas'. Detailed review of the McElhinny et al. (1978) suggest potential problems that could affect the tectonic interpretation, in particular some directions may not have been appropriately corrected for bedding. Six of the 12 sample sites presented by McElhinny et al. (1978) had no bedding correction applied which, considering the proximity of these sites to the significantly tilted beds of our sampled horizons (dip = 33–43°), casts doubt on their reliability.

3.2. Present study

3.2.1. Introduction

The primary objective of our investigation was to gain robust magnetic remanence to constrain the location of the Neotethyan rifting margin between NW Greater India and the adjacent portion of the Cimmerian terrane.

3.2.2. Sampling locations

The main sampling was conducted in Srinagar and the nearby Kashmir Valley, NW India. In addition, material was collected from Sonamarg, ~40 km NE of Srinagar, and the Aharbal Falls area ~60 km south of Srinagar, but they did not yield useful/consistent data, thus these locations will not be discussed further.

3.2.2.1. Locality one: Guryul Ravine and SE Srinagar quarries (Outcrops PT1–PT4). The type locality for the Panjal Traps includes exposures in the Kashmir Valley, specifically those close to Srinagar (Wopfner and Jin, 2009; Huddleston, 1896; Lydekker, 1883). Several quarries near the town of Pampore ~9 km southeast of Srinagar augment the natural exposures. Together, the exposures and workings allow access a quasi-continuous several km-thick, 30–40° dipping stratigraphic succession of Panjal Trap flows and associated strata, from silicic units at the base to the overlying Middle-Upper Permian and Triassic sediments.

Samples were collected from the Guryul Ravine (PT1) and three adjacent quarries (PT2, PT3 and PT4). The total thickness of the basalt-dominated section is ~3300 m. The unit is underlain by ~1300 m of silicic volcanic rocks. Individual flows up to 30 m thick (Chauvet et al., 2008; Shellnutt et al., 2011) dominate the succession. Columnar jointing and the presence of plant fossils in rare interbeds indicate subaerial eruption. Pillow lavas at Guryul Ravine are interpreted to indicate eruption in a lacustrine setting (Nakazawa and Kapoor, 1973). The uppermost flows are brecciated and conformably overlain by the *Gangamopteris* beds, which comprise tuffaceous chert and shale. The basaltic rocks are fresh although some have experienced zeolite facies metamorphism.

Together exposures from outcrops PT1–PT4 form a tectonically continuous and coherent sequence of beds. Thus, we treat the paleomagnetic results from this region as those of a single locality for purposes of determining a formation-mean direction. From analogy to studies of eruption patterns associated with similar LIPs (Renne and Basu, 1991; Richards et al., 1989), we estimate that the ~3 km thick composite section would have been erupted over 1–2 m y. As such this should be sufficient to allow for geomagnetic secular variation to be averaged.

3.2.2.1.1. Outcrop PT1: Guryul Ravine. The Guryul Ravine (074°56.69′ E 34°04.54′ N) is a classic exposure particularly known for Permo–Triassic sediments. It is especially significant for sediments that preserve the biostratigraphical and paleoclimatological signatures characterizing the Permo–Triassic extinction (Sweet, 1970; Matsuda, 1981; Korte et al., 2010). These rocks conformably rest upon the uppermost flows of the Panjal Traps. Critically, the succession demonstrates the nature of the transition from rift to post-rift passive margin regimes that evolved during rifting of eastern Cimmeria from the Indian margin of Gondwana (Gaetani and Garzanti, 1991; Metcalfe, 1994). These upper basaltic flows have been a focus of several studies including (Shellnutt et al., 2011, 2012, 2014, 2015). It is interesting to note that it was not included, or at least not explicitly identified, as a sampling locality in the study by McElhinny et al. (1978). However, it is a possible that the Guryul lavas are correlative of flows at some of their sites e.g. the Atwai Quarries. Strata at Outcrop PT1 dip ~43° toward the ESE (115°). The flows here are 2–3 m thick dark, largely aphanitic, cooling units with tabular boundaries. Primarily basaltic, they were likely to have been erupted sub-aerially (Wopfner and Jin, 2009). There is evidence of epidote alteration, probably penecontemporaneous with eruption and cooling. Some cooling units contain small but abundant veins of quartz and epidote ~5 cm thick; others exhibit 'jigsaw' brecciation. Such flows were avoided as they present difficulties with sample extraction. Four block samples (PT1.1–1.4) were collected from discrete cooling units, and span ~15 m of the stratigraphy.

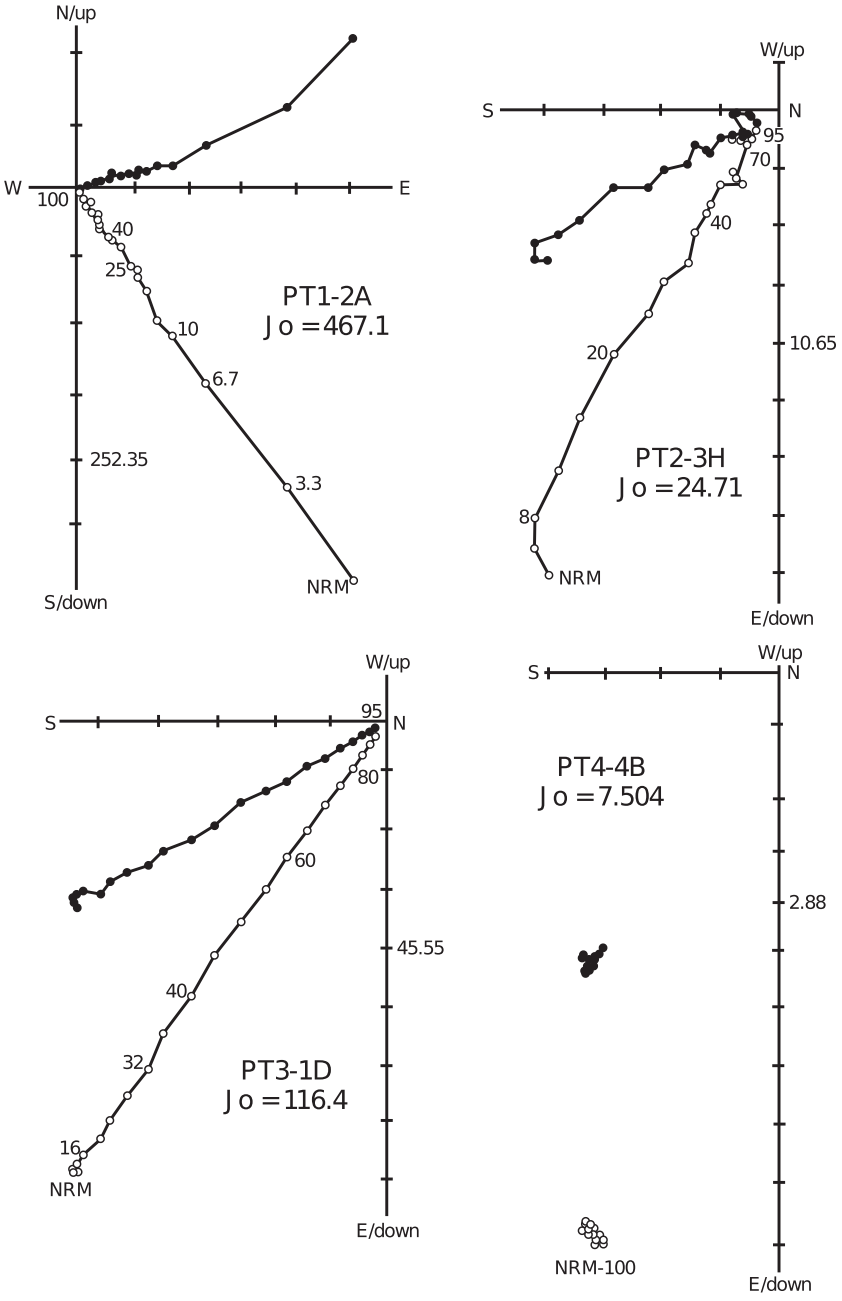


Fig. 2. Examples of AF demagnetization data plotted on vector-endpoint plots (Zijderveld, 1967). In all cases the data are shown in the tilt-corrected reference frame. Solid/open circles represent vectors projected onto the horizontal/vertical planes. For each sample, the identifier and initial natural remanent magnetization intensity (J_{NRM}) are shown (units being mA m^{-1}). Key demagnetization steps are also indicated (up to 100 mT) as well as scale bars on both axes.

3.2.2.1.2. Outcrop PT2: Zewan quarry. This outcrop is a small abandoned quarry SW of Guryul Ravine near Zewan (074°54.72′ E 34°03.07′ N). The bedding attitude is similar to that of Guryul Ravine, that is ~40° to the SE, but the exposure is less extensive. The sequence is estimated to lie ~1 km lower in the stratigraphic succession than that of the Guryul Ravine. Cooling units are typically 2–3 m thick massive basalt flows with tabular boundaries. The lowermost flows are amygdaloidal with epidote infilling. Four block samples (PT2.1, 2.2, 2.3, 2.4) were collected from this site and span ~20 m of stratigraphy.

3.2.2.1.3. Outcrop PT3: Athwajan Quarry. This locality is a large working quarry adjacent to the village of Athwajan, ~5 km west of Zewan (074°52.28′ E 34°02.77′ N). Cooling units are 3–4 m thick flows with tabular boundaries. The overall orientation of the units is ~43° toward the south. Notably the rocks are slightly more felsic than those at in the previous two. Stratigraphically these flows are from the lower part of the Panjal Traps (Shellnutt et al., 2012) approximately 3 km lower in the section than those at Guryul Ravine. Samples were preferentially collected from massive, aphanitic sections of individual cooling units. Although oriented blocks were collected from five flows (PT3.1–3.5) spanning 30–40 m of stratigraphic succession, PT3.4 did not survive the journey to the laboratory and thus could not be analyzed.

3.2.2.1.4. Outcrop PT4: Pantha Chowk Quarry. The locality is a large working quarry near Pantha Chowk (074°53.01′ E 34°02.40′ N) and is close to Outcrop PT3. It was the primary sample site of the Shellnutt et al. (2011) that successfully extracted zircons for radiometric dating (289 ± 3 Ma). Samples from the 'Atwai Quarries near Srinagar town' described by McElhinny et al. (1978) are most likely

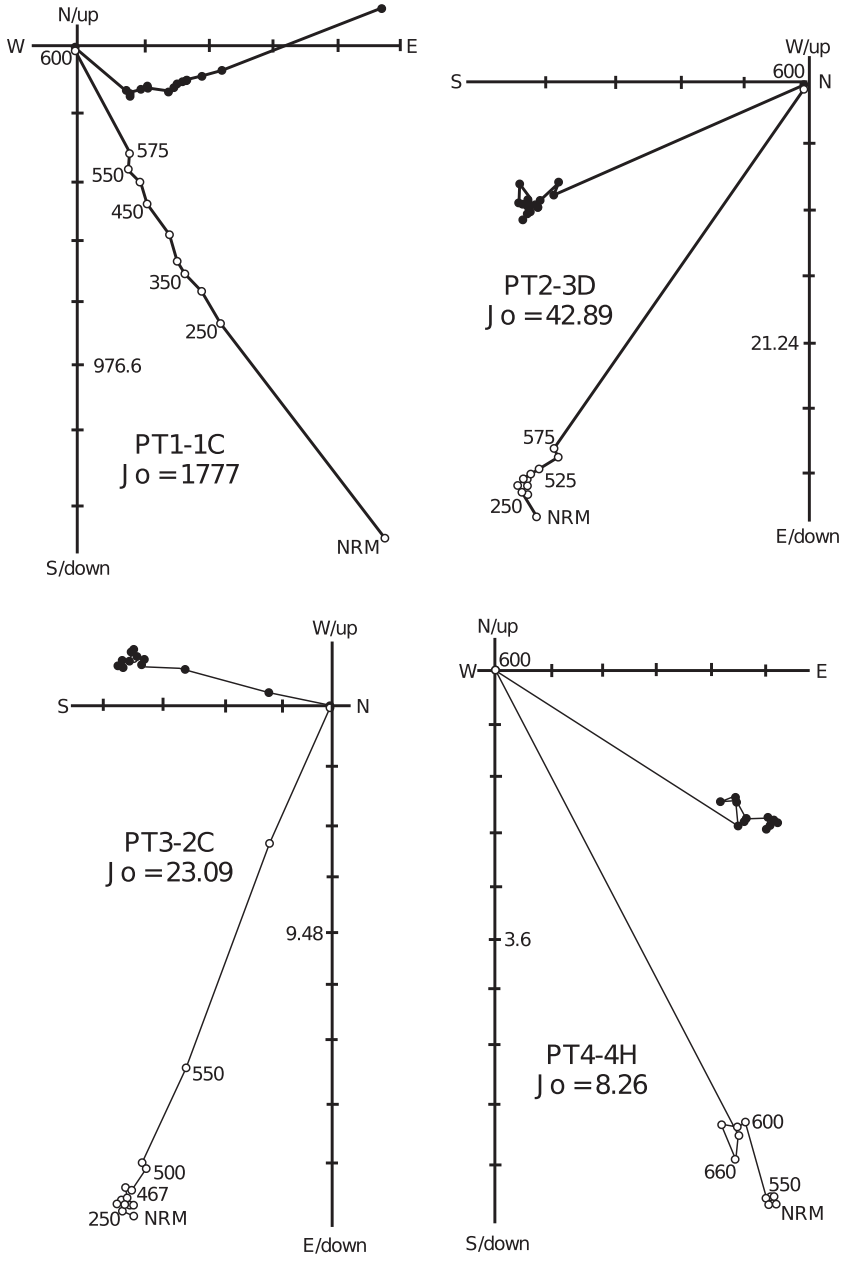


Fig. 3. Examples of thermal demagnetization data plotted on vector-endpoint plots (Zijderveld, 1967). In all cases the data are shown in the tilt-corrected reference frame. Solid/open circles represent vectors projected onto the horizontal/vertical planes. For each sample, the identifier and initial natural remanent magnetization intensity (J_{NRM}) are shown (units being mA m^{-1}). Key demagnetization steps (in °C) are also indicated as well as scale bars on both axes.

from either or both the Athwajan and Pantha Chowk quarries although this cannot be affirmed with certainty. Bedding attitude is 37° to the SE and the characteristically 1–3 m thick cooling units have tabular boundaries similar to those elsewhere. Four block samples were collected at this site (PT 4.1 – PT 4.4) across a 30–40 m of stratigraphy.

3.2.3. Sampling method

At each outcrop oriented blocks (ideally within ±2° in both the horizontal and vertical planes), 500–700 cm^3, were collected from discrete cooling units (all bar one are flows, the other is a gabbroic dyke). Each individual cooling-unit block sample, therefore, represents a paleomagnetic 'site' because the sample should provide an instantaneous record of the geomagnetic field a shortly after the magma had crystallized and the magnetic grains cooled below their blocking temperatures. There is, however, potential for blocks collected in this way to record small sampling-related orientation biases, which is not really a problem with field-based drill-cored collections. Therefore, in an attempt to minimize this effect, we aimed to select reference surfaces with very different orientations to the others in the section. Magnetic bearings were corrected to 'true' using the International Geomagnetic Field Model, version 11; at the time of sampling the offset was ~1°W.

3.2.4. Laboratory methodology

The bulk of the samples were prepared and analyzed at the University of Hong Kong Paleomagnetic Laboratory. Blocks were set in plaster-of-Paris and six to eight 2.54-cm-diameter by 2.15 cm-long core specimens were drilled from each block. A total of 29 block samples were collected yielding ~165 mini-cores of which 126 were analyzed.

The natural remanent magnetization (NRM) of each prepared specimens was measured using an AGICO JR-6A spinner magnetometer. In order to determine the magnetization history of the samples, a stepwise alternating field (AF) demagnetization was carried at steps varying between 2.5 and 10 mT using a "Molspin Tumbler" (which has a peak field of 100 mT). Examples of demagnetization behavior are presented in Fig. 2.

For those specimens that did not respond to the AF treatment (e.g., Fig. 2, Specimen PT4-4B), they were then subjected to thermal demagnetization. This work was carried out using an ASC Scientific (TD 48SC) thermal specimen demagnetizer at the Australian National University RSES Black Mountain Paleomagnetic laboratory. The demagnetization steps were 50 °C between 250 and 400 °C; 33 °C steps from 400 to 500 °C; 20–25 °C beyond 500 °C until total NRM was removed. Examples showing the demagnetization behavior are shown in Fig. 3. Most sample yielded simple magnetization histories, but some displaying 'noisy', erratic results and these were excluded from any analysis. Data were subsequently analysed with software developed by Core Magnetics Ltd., UK and CEMP Systems Pmag Tools 4.2a (Hounslow, 2006).

Characteristic magnetization directions were determined using principal component analysis (Kirschvink, 1980) applied to data plotted on vector-endpoint plots (Zijderveld, 1967). Typically, a stable single direction is observed above 10–20 mT/350–400 °C. Thermal demagnetization indicates these directions are recorded either by magnetite (because they 'unblock' ⩽578 °C, followed by a sharp decline in intensity) or magnetite plus variable fractions of haematite (which unblocks ⩽680 °C). Block-sample- and locality-mean directions were calculated using Fisher (1953) statistics.

We consider the magnetite record to represent a primary signal, whereas any haematite component is an alteration remanance acquired shortly afterwards, very possibly before the next flow.

3.2.5. Results

3.2.5.1. Outcrop PT1: Guryul Ravine. Five mini-cores from each block, with the exception mini-core 4 from PT1.2, were first treated using AF demagnetization. Two additional specimens from each block were subsequently subjected to thermal demagnetization. NRM intensities ranged from 240 to 2100 mA/m. in samples PT1.1, 1.2, 1.3, a viscous remanence was removed at ⩽8 mT. Beyond this, samples carried a single-component magnetization with median destructive fields between 6 and 10 mT with a reversed polarity. All PT1 samples reached unblocking temperatures between 550 °C and 600 °C suggesting that the remanence is carried by magnetite. While characteristic magnetization directions were identified for the PT1.4 specimens, the scatter in directions was deemed too large ($\alpha 95 = 22.0°$, $k = 10.0$) thus the mean direction was not used in calculating the exposure mean.

3.2.5.2. Outcrop PT2: Zewan Quarry. Five mini-core specimens from blocks PT 2.1 and 2.3 and four from blocks PT 2.2 and 2.4 were treated initially using AF demagnetization. As a consequence of two specimens each from PT 2.1 and 2.3 and one specimen from PT 2.2 and 2.4 not responding to the treatment, they were later subjected to thermal demagnetization where unblocking was achieved at 550–600 °C. NRM intensities ranged from relatively low in PT2.1 (7.3–9 mA/m) to very high 160–940 mA/m in PT2.2 with intermediate values in PT2.3 (13–33 mA/m) and PT2.4 (13–52 mA/m). The large range in NRM suggests varying concentrations of magnetic minerals but consistent directions were recorded nonetheless with median destructive fields between 6 and 8 mT. Although plots for mini-core specimens from PT2.1 are 'noisy' due to low magnetic intensities, consistent single-component remanences with reversed polarity were isolated in all block samples.

3.2.5.3. Outcrop PT3: Athwajan Quarry. Five specimens from each block sample, excluding PT3.5, were AF demagnetized. Two additional specimens (only one from PT3.1) from each block were subsequently subjected to thermal demagnetization where unblocking occurred at 550–600 °C. In PT3.1, 3.2, 3.3, NRM intensities ranged from 1.3 to 120 mA/m. Pilot specimens from PT 3.5 returned intensities so low (0.01–0.73 mA/m) as to be out of the reliable range of the magnetometer. Consequently, ChRMs could be isolated and specimens from this block were not tested beyond the pilot measurements. From the remaining samples, specimen analysis plots indicating a prominent single magnetization component with a reversed polarity. A single mini-core specimen from PT3.3 (PT3.3b) returned an anomalous direction (possibly due to a sampling error) and was excluded from the sample mean calculation.

3.2.5.4. Outcrop Pt4: Pantha Chowk Quarry. Five specimens drilled from blocks PT4.1 and 4.2 were satisfactorily demagnetized using AF with two additional mini-core specimens subjected to thermal demagnetization, again with unblocking at 550–600 °C. The pilot samples from PT 4.3 and 4.4 did not respond to the treatment and thus all of the cores from these blocks were subjected to thermal demagnetization. Unsurprisingly, these all unblocked at 680 °C and 700 °C (indicative of a hematite carrier). NRM intensities ranged from 0.7 to 17.7 mA/m. Despite 'noisy' plots due to low intensities, specimens from PT4.1 and 4.2 yielded single component, reverse-polarity remanences. The specimens from PT 4.3 and 4.4 also bore single directions similar to those of PT4.1 and 4.2.

3.2.6. Calculating a formation mean direction

Treating the four localities as (PT1–4) as a tectonically coherent section, a Fisher mean direction calculation for all samples, excluding PT 1.4 and 3.5, gives a locality mean of Dec: 134.8°, Inc: 55.3°, where $\alpha 95 = 8.9°$ and $k = 21.0$ with $n = 14$; Fig. 4). An inclination

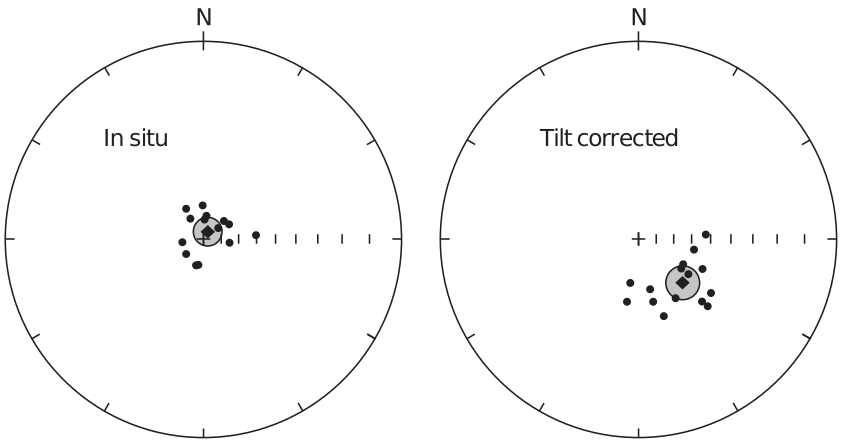


Fig. 4. Equal angle stereonets summarizing the mean directions of the fourteen samples sites in locality 1 used to calculate the locality mean (shaded circle). (A) *In situ* Dec: 29.0°, Inc: 85.1° (α95 = 8.2°, k = 24.6, n = 14) and (B) tilt corrected Dec: 134.8°, Inc: 55.3° (α95 = 8.9°, k = 21.0, n = 14). The α95 values for each of the block samples are listed in Table 1; range 0.9–12.5°.

Table 1
Summary table of paleomagnetic results for Panjal Traps samples. Specimens are listed for each outcrop and related to individual blocks collected from those outcrops and the core samples drilled from each block. Nc(Np)Nt: Nc denotes number of specimens used to calculate a sample mean out of number of specimens processed Np. Nt is the number of Th demagnetizations (inclusive). MAD = maximum angular deviation.

Specimen	Strike/Dip/DD	Nc(Np)Nt	*in situ*		Tilt Corrected		α95	k	Polarity	Used in mean calculation
			Dec	Inc	Dec	Inc				
Locality 1: Guryul Ravine and SE Srinagar quarries										
Outcrop PT1: Guryul Ravine (074°56.69′ E 34°04.54′ N)										
PT1-1	025/43ESE	6(7)2	326.5	75.9	101.0	58.3	6.6	105.4	R	Y
PT1-2	025/43ESE	4(6)2	358.2	70.8	86.2	52.4	9.5	95.3	R	Y
PT1-2	025/43ESE	7(7)2	191.5	75.0	134.7	41.7	3.0	404.2	R	Y
~~PT1-4~~	~~025/43ESE~~	~~6(7)2~~	~~96.9~~	~~28.0~~	~~300.0~~	~~70.9~~	~~17.1~~	~~21.0~~	~~R~~	~~N~~
Outcrop PT2: Zewan Quarry (074°54.72′ E 34°03.07′ N)										
PT2-1	054/33SE	7(7)2	229.8	76.7	166.9	53.8	5.7	112.6	R	Y
PT2-2	054/33SE	5(5)1	261.5	77.7	166.8	60.8	12.5	38.3	R	Y
PT2-3	054/33SE	7(7)2	196.0	74.2	161.9	45.7	9.1	45.2	R	Y
PT2-4	054/33SE	5(5)1	53.1	79.5	125.1	55.7	4.4	308.5	R	Y
Outcrop PT3: Athwajan Quarry (074°52.28′ E 34°02.77′ N)										
PT3-1	083/43S	6(6)1	59.7	73.2	148.0	51.1	0.9	5920.8	R	Y
PT3-2	083/43S	7(7)2	330.3	70.0	190.9	64.5	2.5	603.4	R	Y
PT3-3	083/43S	7(7)2	85.8	60.4	134.1	38.3	9.7	39.4	R	Y
Outcrop PT4: Pantha Chowk Quarry (074°53.01′ E34°02.40′ N)										
PT4-1	050/37SE	6(6)1	48.4	74.4	114.9	50.7	3.1	481.8	R	Y
PT4-2	050/37SE	7(7)2	97.8	75.0	126.7	41.0	3.5	306.6	R	Y
PT4-3	050/37SE	7(7)7	2.1	78.6	124.3	60.6	4.9	153.5	R	Y
PT4-4	050/37SE	7(7)7	6.9	76.4	119.4	60.8	2.4	620.6	R	Y
Mean of 14 from 15 samples *in situ*		14(15)	29.0	85.1	–	–	8.2	24.6	R	
Mean of 14 from 15 samples tilt corrected		14(15)	–	–	133.8	55.8	8.9	21.0	R	
Mean inclination only tilt corrected		14(15)	–	–	–	52.5	4.9	47.2		

only mean, using the statistics of McFadden and Reid (1982), gives Inc = 52.5°, where α95 = 4.9° and k = 47.2 with n = 14).

3.2.7. Discussion

The single-component nature of these fourteen PT1–4 outcrop flows leaves two avenues open for interpretation: (1) the magnetization is primary and that this particular exposure of the Panjal Traps, in contrast to the other outcrops in Kashmir has been spared from alteration and overprinting; (2) the original magnetization has been completely overprinted throughout the entire block during a later event as suggested by McElhinny et al. (1978).

Consistency of results is demonstrated with general agreement in results over both a significant stratigraphic and areal range. Although the amount of spread between results through the stratigraphy is appreciable it is not unacceptable. Indeed the directional scatter suggests that magnetizations were acquired and different times by successive flows and record fluctuations in the geomagnetic field over the eruption period spanning a few million years, and are thus primary. Critically, a tighter grouping would suggest a much younger, 'wholesale' overprint acquired simultaneously at all depths over a relatively short period. A primary magnetization scenario thus seems to be the most parsimonious interpretation. The tilt-corrected mean is (Dec: 134.8°, Inc: 55.3°, α95 = 8.9°, k = 21.0, n = 14) yields a paleopole (south) of 110.5°E 8.4°S (A95 = 10.7). As paleolatitude is derived from inclination alone, the inclination-only mean is 52.5° (α95 = 4.9, k = 47.2,

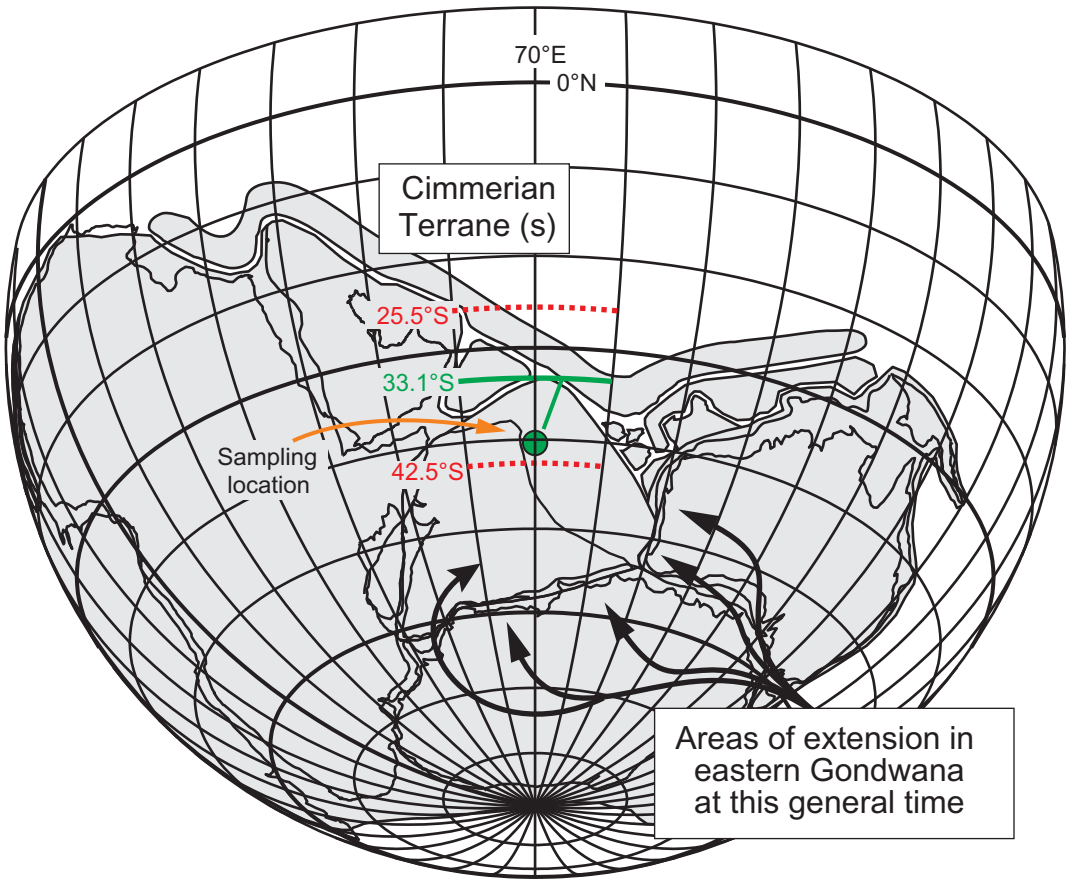


Fig. 5. Gondwana reconstructed to in the late Early Permian (280 Ma) using rotations from Torsvik et al. (2008) after Ali et al. (2012). Extrusion location of the Panjal Traps is plotted according to the paleolatitude constraint calculated in this study (33°S ± 4.9°) and in accordance with the inferred configuration of the eastern Cimmerian terranes. The sample site (locality 1) is plotted at corresponding present-day coordinates on cratonic India. The disparity between the two was calculated to be ~8°.

n = 14) (McFadden and Reid, 1982). Assuming that the signal is primary and given the Early Permian Gondwana context of the Panjal Traps, this yields a paleolatitude of 33.1 ± 5° S.

Considering northern India's position in the mid-latitudes of the Southern Hemisphere during the Permian, the magnetization is interpreted to be of reversed polarity. Therefore the tilt-corrected mean direction cannot be a recent acquisition as it is not consistent with the present field (Tauxe, 2010). Consistency between varying NRM intensities and mineralogies, magnetite and hematite, suggests that the hematite formed shortly after crystallization. Usually, a primary magnetization plot will show significantly tighter grouping after bedding correction. However, in this case the reduction in scatter is small.

3.2.7.1. Comparison to previous studies. Given the significant contrast of these new data with the earlier study of McElhinny et al. (1978) comparison between the two is warranted. It may be noted that the new and old datasets both plot with the same polarity and in the same quadrant. Indeed, the mean declination varies by only 16°. The main discrepancy is in the inclination means with those reported by McElhinny et al. (1978) being significantly shallower at 32.2° ($\alpha 95 = 9.0°$, $k = 17$). This inclination corresponds to a paleolatitude of ~17.5°, thus the suggestion that magnetization was a 'Tertiary overprint'.

Interestingly, McElhinny et al. (1978) report a positive fold test but the location of the folded strata used is not given. As such structures were not observed at the sample sites visited in this study, it is surmised that at least some of the sample sites used in their study have a distinctly different structure to the Guryul volcanic pile. While inconsistencies in the earlier dataset have been approached forensically, inevitably some degree of speculation may be introduced. One suspected "problem" is that the bedding corrections were not fully applied in the earlier work. Whatever the explanation, this is ultimately impossible to determine without the essential basic geological data of their sampling sites, which the 1978 publication lacks.

3.3. Implications

The orientation of magnetization in the Panjal Traps is consistent with the Sakmarian/Artinskian age of the unit which is well constrained both stratigraphically (Wopfner and Jin, 2009) and radiometrically (289 ± 3 Ma) (Shellnutt et al., 2011). Importantly, the reversed polarity of the magnetization is consistent with the accepted geomagnetic orientation for this age corresponding to the ~318–262 Ma Kiaman Reverse Polarity Superchron (Opdyke et al., 2000; Cottrell et al., 2008; Irving and Parry, 1963).

The inclination-only mean of magnetization, 52.5° ($\alpha 95 = 4.9°$), indicates eruption occurred at mid-latitudes ~33.1°S (error range is 42.5°S to 25.5°S) (Fig. 5). The Panjal Traps thus erupted during rifting of the Qiangtang and western Lhasa terranes at this latitude. This constrains the position of the relevant portion of the Early Permian Neotethyan margin, and by inference fixes the maximum extent of NW Greater India. These data are consistent with eastern Cimmerian configurations inferred from the paleomagnetic data of the Baoshan terrane (Ali et al., 2013). The direct association between extrusion of the Panjal Traps and the exact configuration of the Qiangtang, Sibumasu and Lhasa terranes in this region is yet

to be established. Basalts in the Karakoram postulated to be related to this rifting may provide such a point of correlation (Rao and Rai, 2007), however, their equivalence to the Panjal Traps is yet to be determined. Using a global reference frame model (Torsvik et al., 2008) reconstruction back to the early Permian (~280 Ma) there is ~8° of separation between cratonic India and the inferred site the Panjal Traps rifting center constrained to 33°S.

With regard to the extent of Greater India, the review paper of Ali and Aitchison (2005) suggested that the extension to the NW of the craton was ~900 km. This is somewhat larger than estimates for the Western Syntaxis area of ~600 km given by models using the Wallaby–Zenith constraint but is consistent with the estimate of ~950 km for the central areas (Ali and Aitchison, 2005, 2006; Gibbons et al., 2012). Assuming that main eruptive center for the Panjal Traps is within the Kashmir Valley, somewhat east of the Western Syntaxis, and further south relative the IYTS, this discrepancy may be accommodated. Notably, the possible ~10° range indicated by the error probability ellipse (± 4.9°) introduces an uncertainty up to 1100 km. Necessarily this places significant limitations on the conclusions we may draw solely from these data. Thus our inferences remain indicative rather than conclusive.

3.4. Comparison to other Late Paleozoic poles for India

Direct comparison with other Permian Indian poles is difficult, if not impossible, due to a lack of reliable data for this region and period. Several studies report Late Paleozoic poles from Himalayan and Peninsula India, the majority dating to the 1970s and 1980s (e.g. Klootwijk, 1975; Klootwijk et al., 1983). However, as none are included within the IAGA Global Paleomagnetic Database (Pisarevsky, 2005) this suggests limited reliability for these results.

The only pole currently listed in the IAGA database and in use in modern reconstructive tectonic references frames (Torsvik et al., 2008, 2012; Van der Voo, 1993) is that derived from the Alozai Formation of Baluchistan, NE Pakistan (111°E 18.1°N) (Klootwijk et al., 1981). Besides being of questionable reliability ($\alpha 95 = 12.1°$ but with $k = 6.0$), deeper investigation of the provenance of the paleomagnetic data shows that the assumed ~280 Ma age (271–318 Ma in the IAGA Database, 289.5 Ma in Torsvik et al. (2008)) of the unit was poorly constrained at the time of that particular paleomagnetic investigation, having been placed within a very broad range of 'Permo-Carboniferous to Liassic' (Klootwijk et al., 1981). More recently, the Alozai Group has been reassigned with some confidence to a Middle and Lower Jurassic age (Maldonado et al., 2011). This pole is thus rejected as representative of the Early Permian. Consequently, a pole derived from the new Panjal Traps data might thus be said to represent the only robust paleomagnetic dataset from northern India of Late Paleozoic age.

4. Conclusions

Tectonic reconstructions of Gondwana back to the Early Permian are constrained using global hybrid data with the assumption that relationships between South Africa, Madagascar, and India are fixed during this time (Torsvik et al., 2012). The relevance of our new Early Permian paleomagnetic data for India lies in their potential to constrain the breakaway Cimmerian components of the Neotethyan margin. Our study highlights the potential for key geological constraints such as age and structural data to be overlooked when dealing with the large, global dataset allowing the incorporation of erroneous results. Conversely, there is also potential for immediately relevant data or resources to be dismissed. The new, primary paleomagnetic data from the Panjal Traps reported here represents a contribution to the larger Indian and, indeed, global paleomagnetic dataset filling a gap in the record for Early Permian India and Gondwana.

The results of this study and its conclusions are bound by significant limitations due to the small sample size arising from local insurgency that made circumstances difficult for fieldwork at the time. Nevertheless, the data are indicative of the potential of the Panjal Traps, and other Cimmerian rift basalt sequences, to yield paleomagnetic data that have significant implications for the nature of the Greater Indian Neotethyan margin and its evolution through time.

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